Can Parallel-Scalable RDBMSs Break the Downsizing Logjam?

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Agenda

- Facing the downsizing logjam
- Parallel-scalable relational databases
- Benefits
- What to look for
- Wrap-up

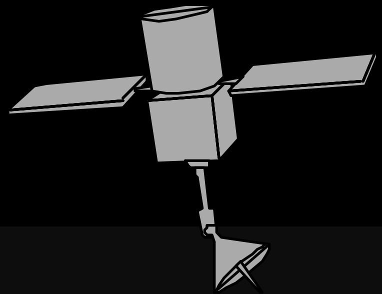
Parallel-Scalable RDBMS Benefits

- Improved performance can be a hardware-upgrade equivalent
- Increased robustness, availability
- Lower-cost configuration flexibility
- Improved scalability curve

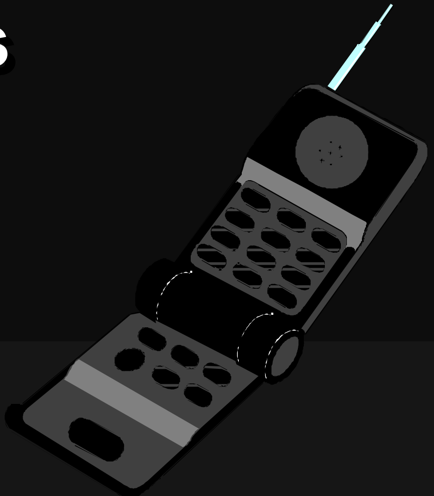
Downsizing Logjam

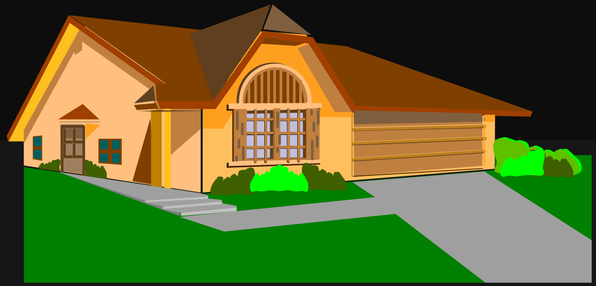
- Driving forces
- Good old days are gone
- Information logs create a jam
- Red summer dresses
- Why empower users?

Driving Forces

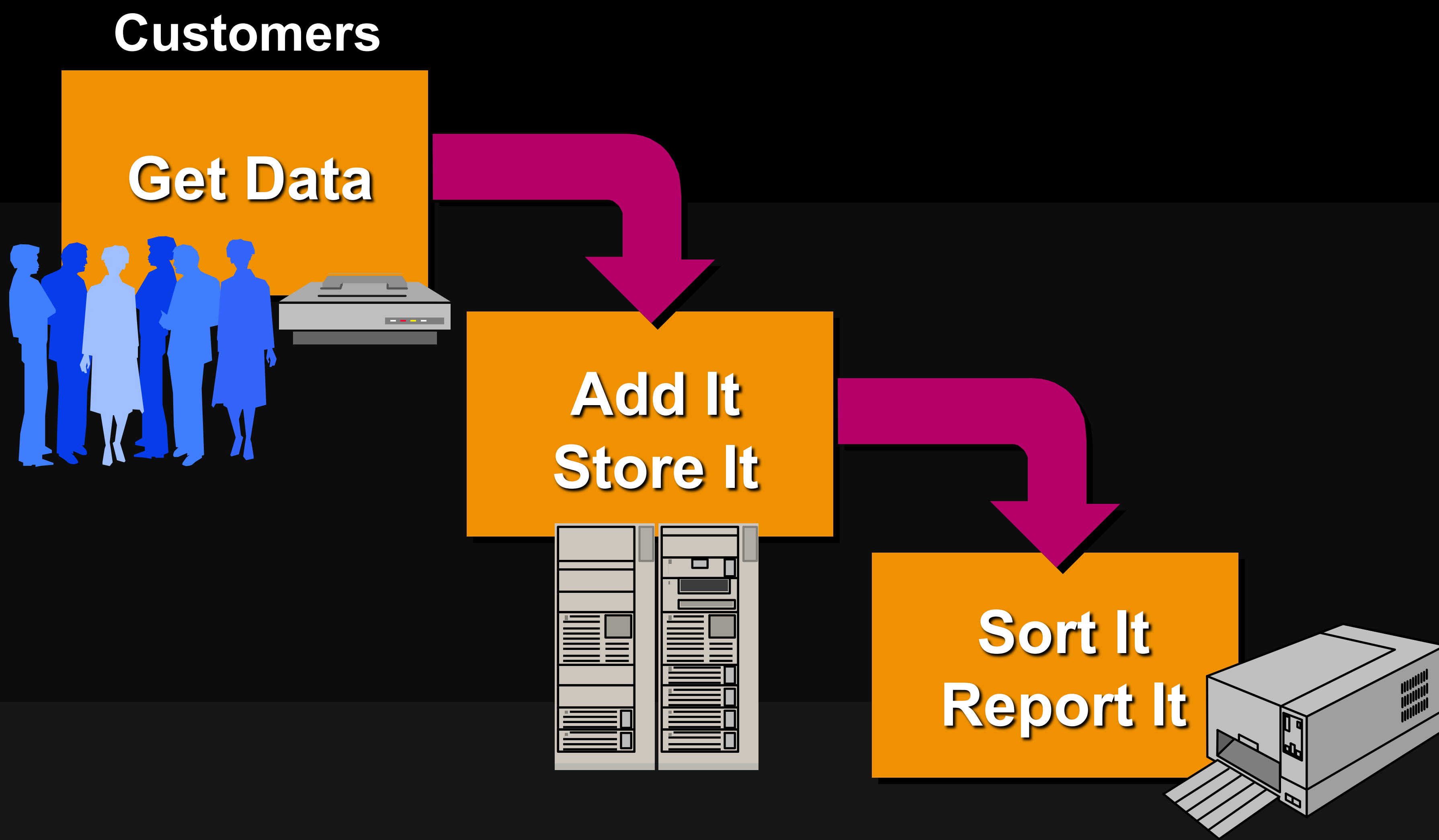


- Global competition
- Technology innovations
- Social changes





Good Old Days are Gone



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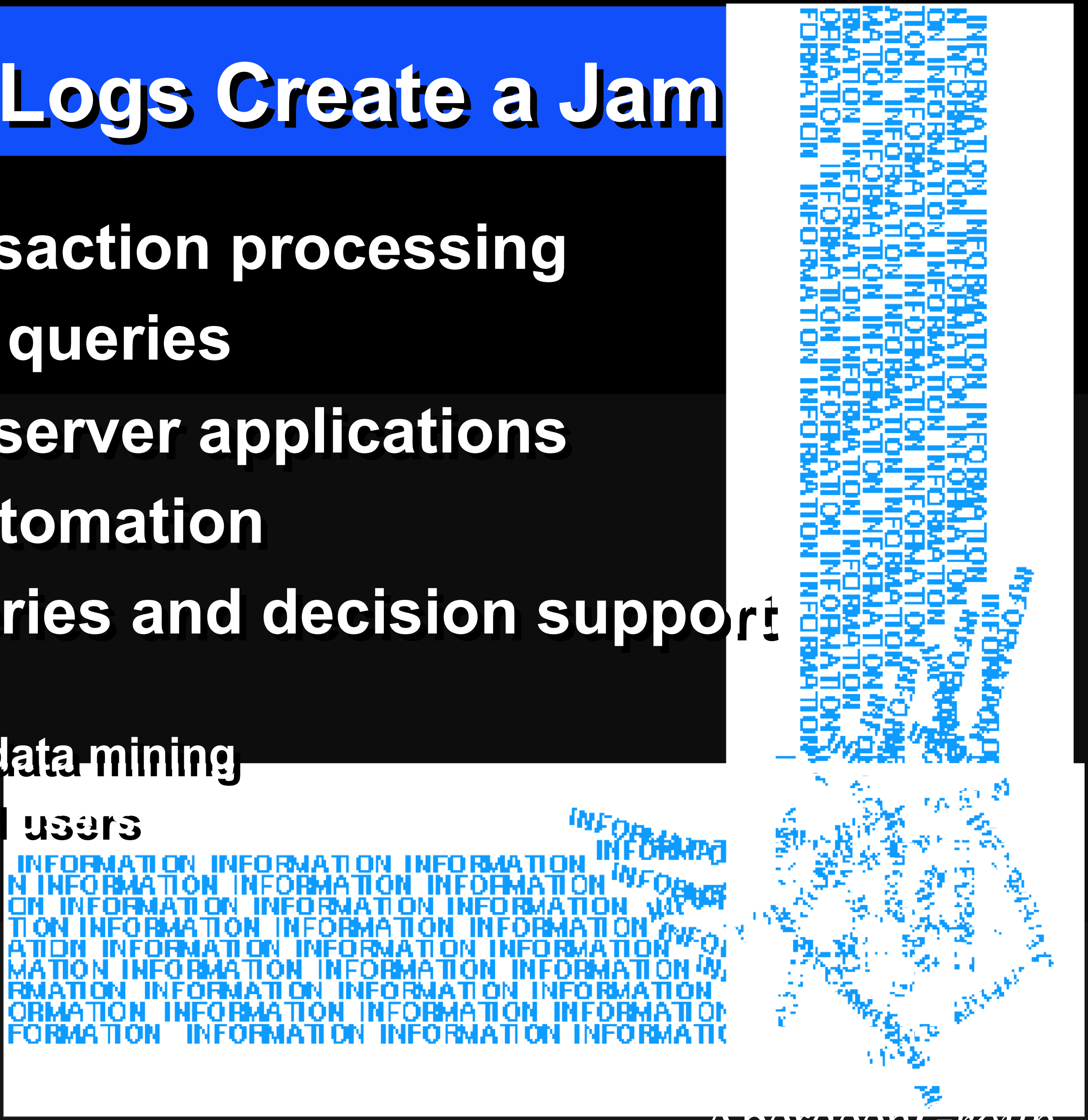
Information Logs Create a Jam

- Online transaction processing
- Production queries
- New client-server applications
- Desktop automation
- Ad hoc queries and decision support
  - EIS
  - Enterprise data mining
  - Empowered users



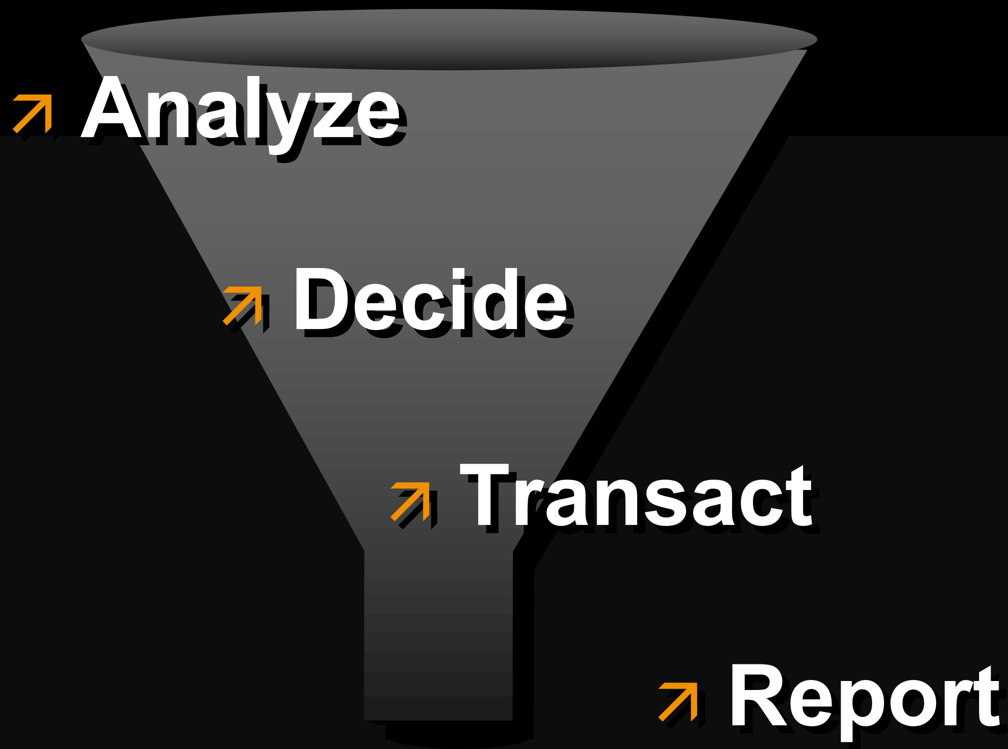
Red Summer Dresses

- It's May in Miami. Your store manager has 1,000 red summer dresses that have not sold. Does the company:
  - Put them on sale?
  - Ship them to Boston?
- Be careful. The correct decision drops directly to the bottom line!

Why Empower Users?

- Personnel downsizing is a reality
- Computer literacy improving
- Increased job-level productivity
- Users are decision makers
- New decision-making cycle

New End User Decision-Making Model



Information Demands Go Through the Roof

- Online transaction processing
- Production queries
- New client-server applications
- Desktop automation
- Ad hoc queries and decision support

3-5x increase in information demand by late 1990s

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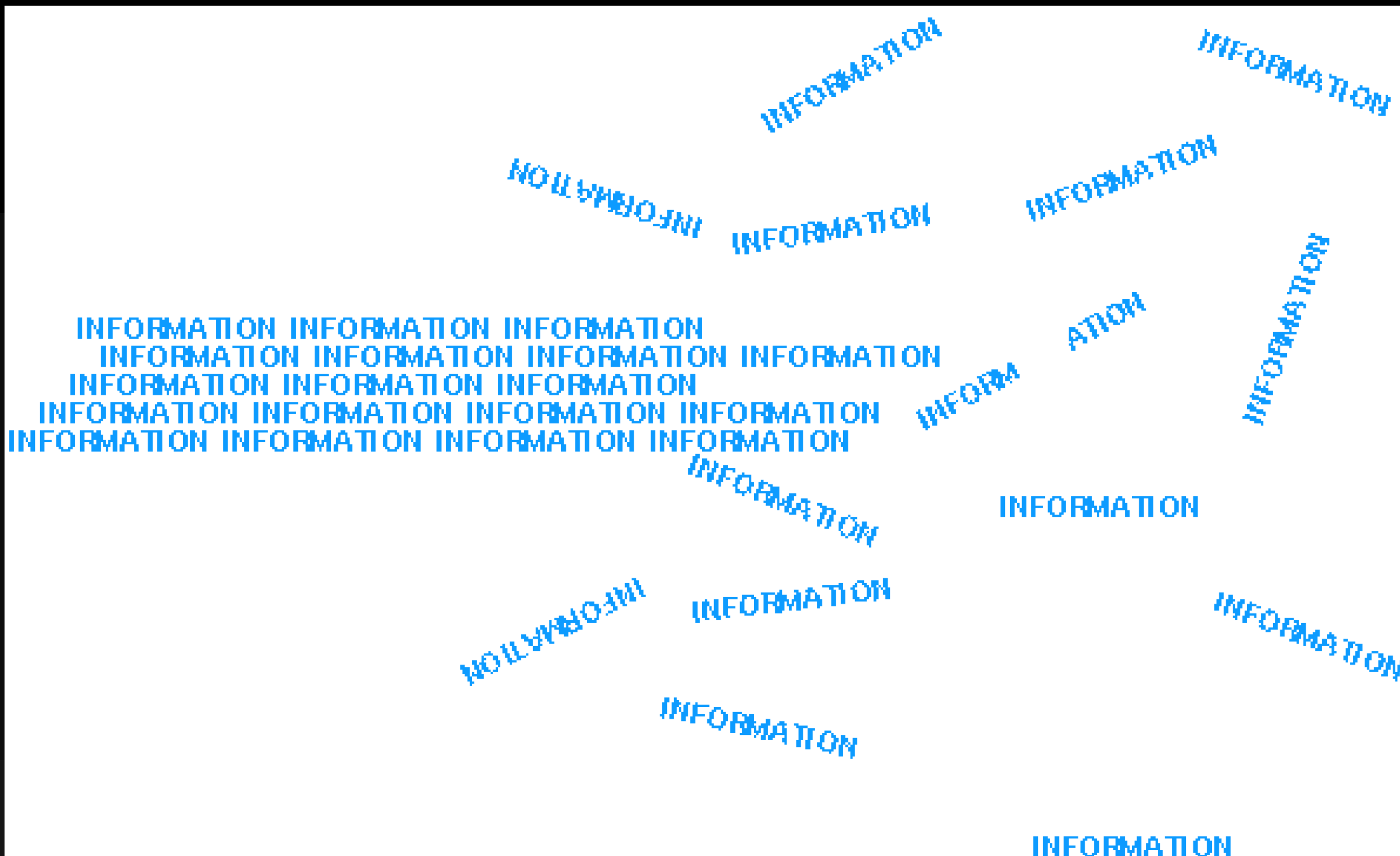
IS on a Hot Tin Roof

- CEO pressure to reduce enterprise costs
- Users demand more and better information
- New client-server applications
- Keeping up with the competition
- Don't forget legacy system migration

Breaking the Logjam

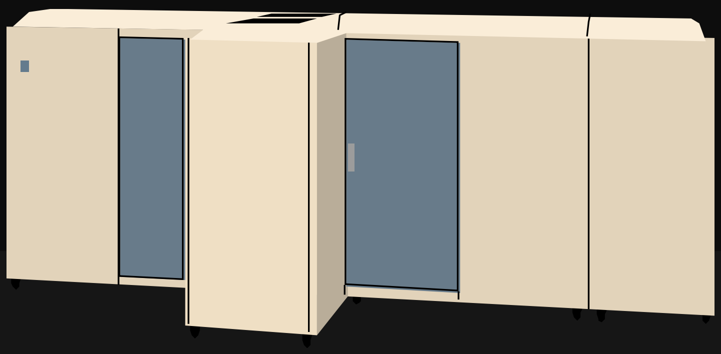


Technology Goals

- Excellent cost-performance
- Scalability
- Availability
- Fast response times
- Open infrastructure
- Development tools
- Ease of administration

Midrange Servers Win

- Excellent cost-performace
- Scalability beyond mainframes
- High availability with clusters
- Fast response times
- Support for open standards
- Widely available development tools
- Open systems management



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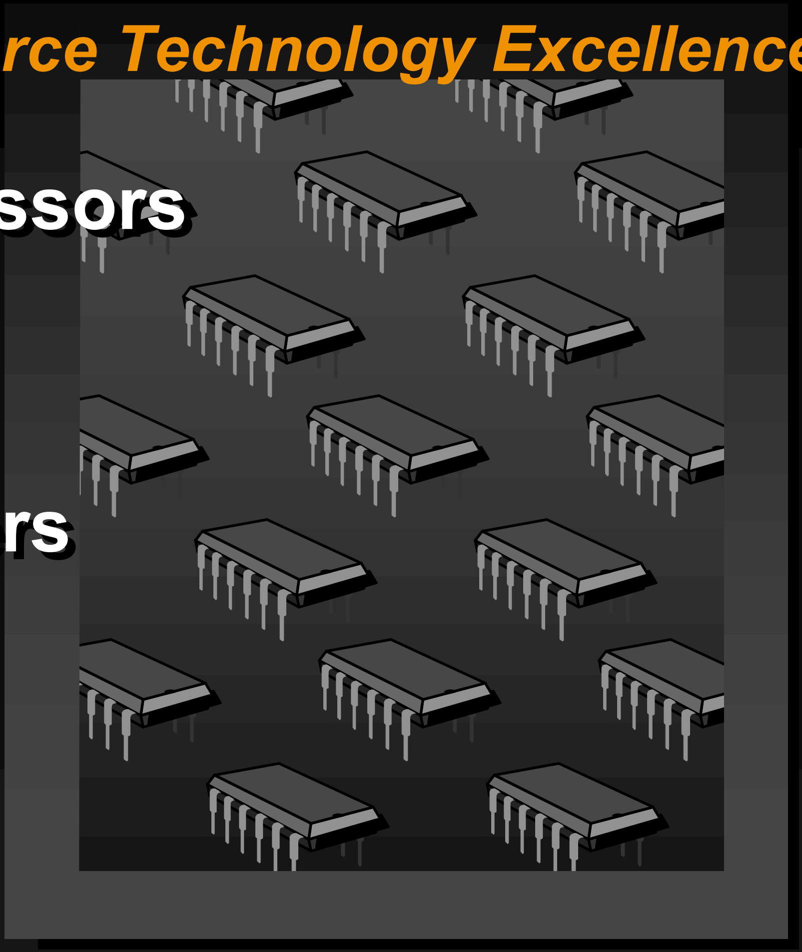
Midrange Server Technology Ingredients

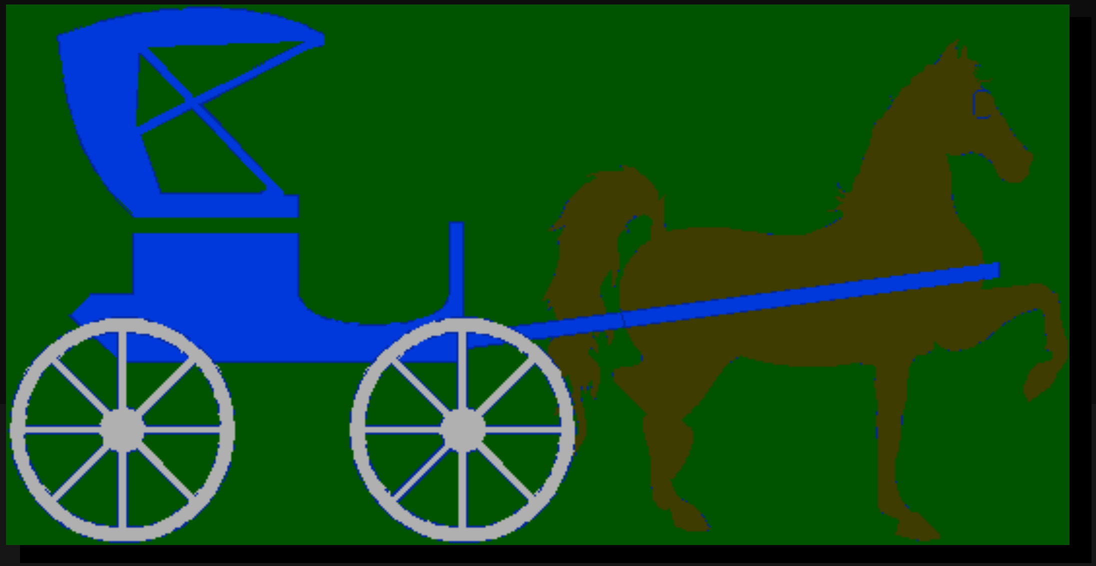
Competitive Pressures Force Technology Excellence

- Lots of microprocessors
- Lots of memory
- Fast I/O buses
- Intelligent controllers
- Unix-on-steroids



Relational Databases are the Biggest Problem

- Coarse-grain multiprocessor support
- Administration bottlenecks
- Poor scaleup in database size
- Few ways to speed up response times



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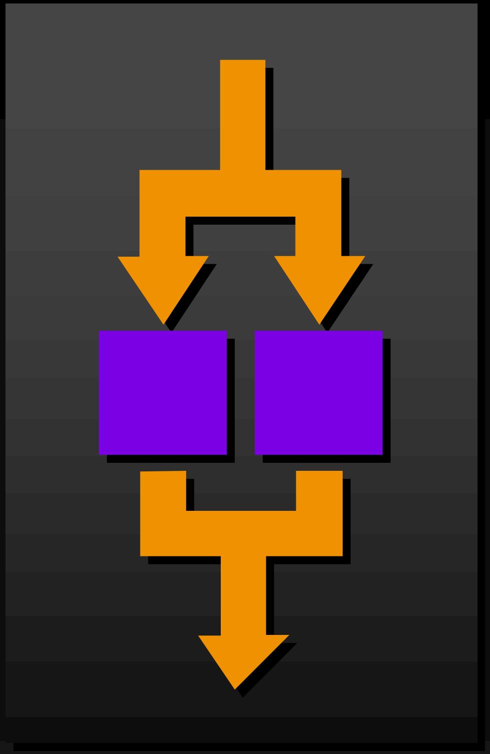
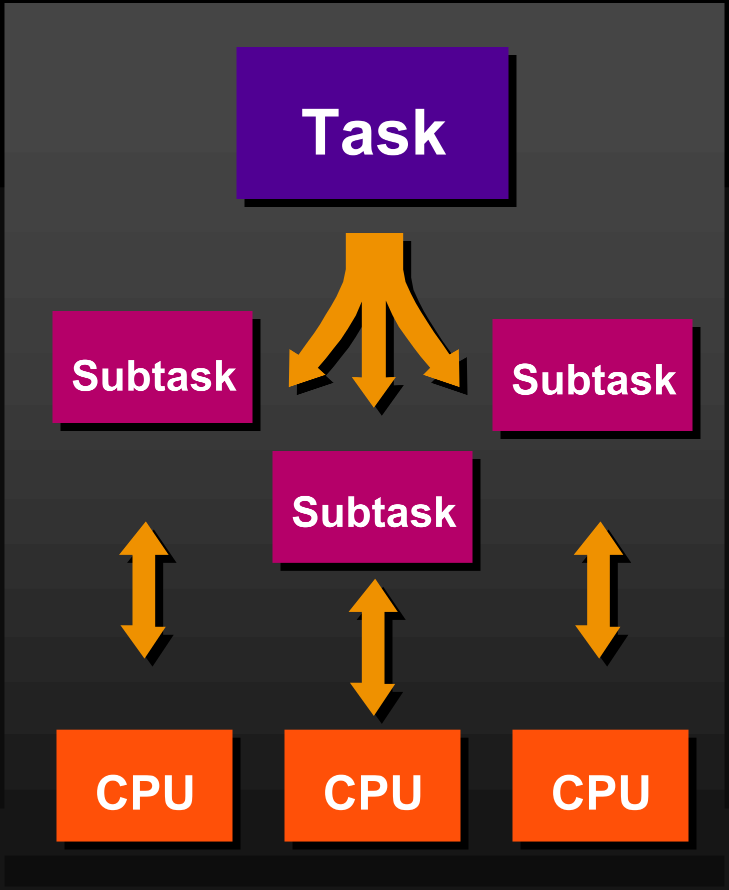
Parallelizing the RDBMS Solves the Problem

- Fine-grain multiprocessor support
- Parallel administration
- Dynamic resource scalability
- Speed-up batch by parallelization



Fine-grain Multiprocessor Support

- Harness available CPUs
- Sort, index, aggregate, scan, merge operations
- Application scalability
  - SMP
  - Clusters
  - Massively parallel

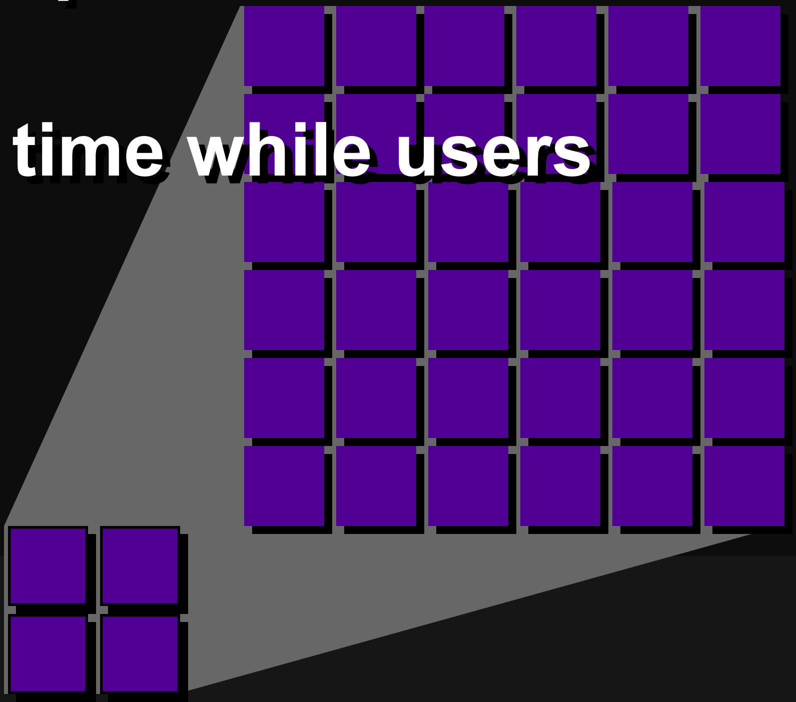


Parallel Administration

- Bulk loading/unloading
- Backup/recovery
- Index building
- Mass updates/deletes
- Alter/reorg

Dynamic Resource Scalability

- Add processors ...
- Add users while response time remains constant
- Improve response time while users remain constant



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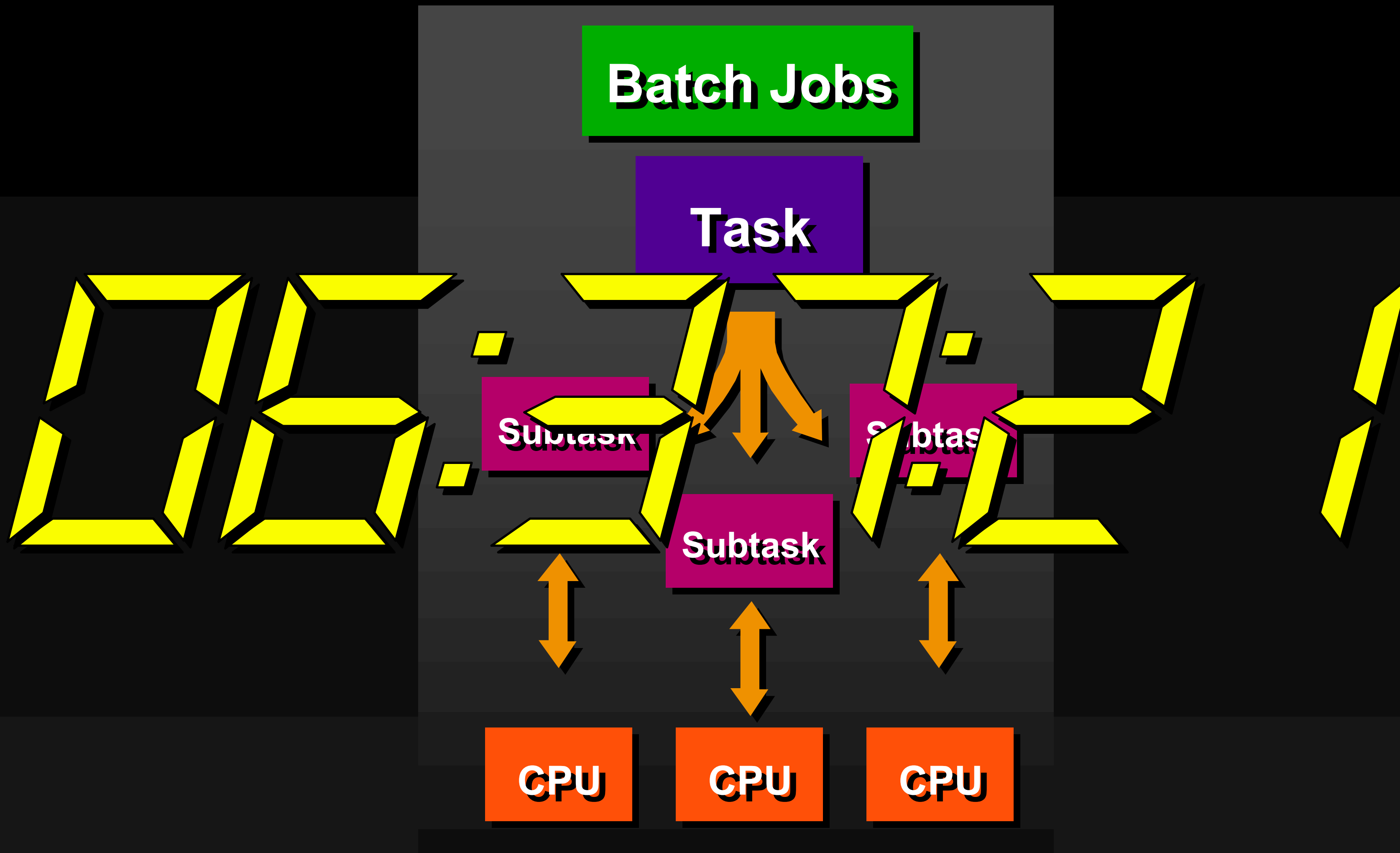
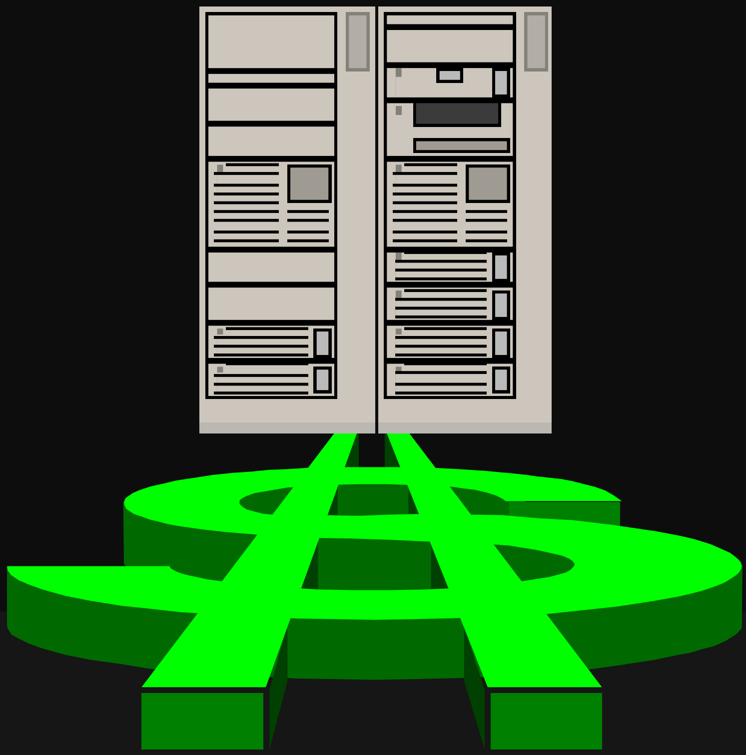
Speed-up Batch by Parallelization



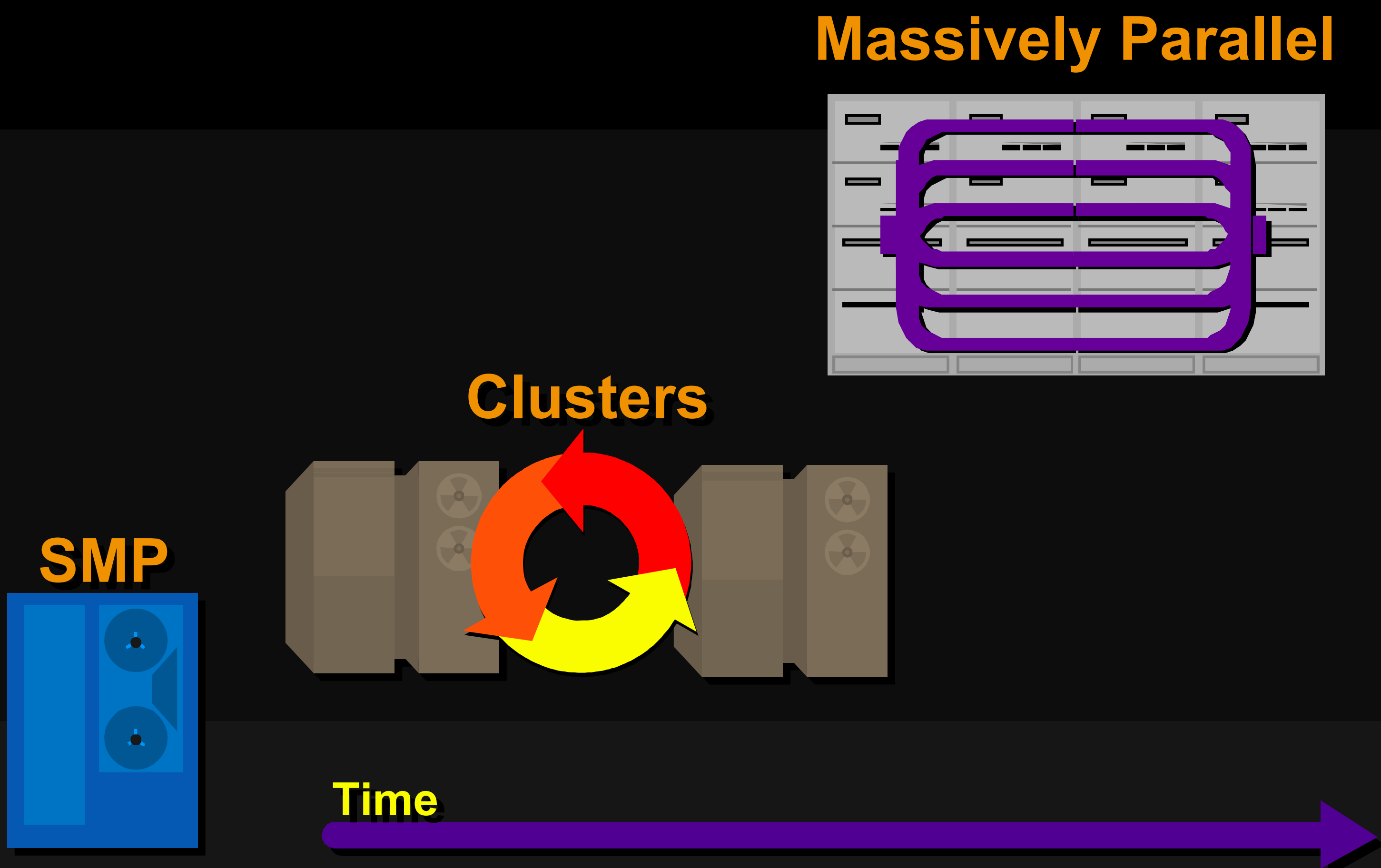
Harnessing Today's SMP Servers

Parallel-scalable RDBMS is like a free hardware upgrade

- SMP is today's most prevalent hardware architecture
- Hardware needs change dynamically
- Buyer's:
  - more users on same hardware
  - faster response times
  - smaller, lower-cost initial hardware investment



Clusters and Massively-Parallel Processors Tomorrow



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What to Look for in Parallel-Scalable RDBMSs

- Hardware adaptation: SMP, cluster, MPP
- Parallelization breadth and depth
- Benchmarks
- Administration
- Technology invisibility

Parallel-Scalable RDBMS Benefits

- Improved performance can be a hardware-upgrade equivalent
- Increased robustness, availability
- Lower-cost configuration flexibility
- Improved scalability curve

Aberdeen Conclusions

- Information-hungry users and applications demand better data management approaches
- Technology pieces now fit together
- Parallel-scalable RDBMS technology
- Solves real-world problems
- Is now being rolled out by RDBMS leaders

Aberdeen Conclusions

Parallel-Scalable RDBMS technology

- Smashing the logjam, empowering users
- SMP now, clusters and MPP soon

IS should take a hard look now at parallel-scalable solutions